

KASTSYUKOVICHY, Belarus

WMO: 268870

Lat: 53.3456N

Lon: 32.0738E

Elev: 549

StdP: 14.41

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.7	-2.0	-12.7	2.8	-6.4	-7.3	3.8	-1.4	25.1	25.8	22.3	27.5	5.0	70	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	85.0	68.6	81.5	66.8	78.5	65.4	70.4	81.2	68.6	78.8	67.0	76.4	8.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.6	99.8	75.9	64.9	94.2	73.6	63.4	89.0	71.8	34.6	81.5	33.1	79.0	31.8	76.5	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.4	19.6	17.2	-15.1	89.0	4.9	4.4	-18.6	92.2	-21.4	94.7	-24.1	97.2	-27.6	100.4		
(5)			-15.4	73.5	4.8	3.1	-18.9	75.8	-21.7	77.6	-24.4	79.4	-27.8	81.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.9	21.7	22.8	31.8	45.9	55.6	62.0	66.4	63.6	54.0	43.2	33.6	25.3		
	DBStd	17.89	11.14	11.45	8.47	7.78	7.93	6.23	5.49	6.11	6.54	7.29	9.14	10.66		
	HDD50	3935	876	762	565	169	34	1	0	0	34	234	493	767		
	HDD65	7893	1341	1182	1029	572	305	132	48	101	333	676	942	1232		
	CDD50	1724	0	0	1	47	208	360	509	421	154	23	1	0		
	CDD65	207	0	0	0	0	14	41	92	57	3	0	0	0		
	CDH74	1945	0	0	0	7	144	339	822	583	48	2	0	0		
	CDH80	493	0	0	0	1	19	51	228	191	3	0	0	0		
	WSAvg	7.8	8.3	8.3	8.7	8.2	7.9	7.2	6.5	6.8	6.9	8.0	8.2	8.0		
	PrecAvg	25.6	1.5	1.5	1.4	1.6	2.5	3.0	3.3	2.4	2.4	2.5	1.9	1.7		
	PrecMax	33.2	3.0	2.8	2.8	3.8	5.2	6.7	9.5	6.1	5.2	5.4	3.5	3.6		
	PrecMin	18.0	0.4	0.6	0.2	0.5	0.6	0.7	0.8	0.4	0.3	0.3	0.2	0.6		
	PrecStd	3.9	0.7	0.6	0.6	0.8	0.9	1.3	1.9	1.3	1.2	1.5	0.9	0.8		
	DB	42.9	43.1	58.4	73.5	82.6	84.6	90.6	95.0	79.5	68.9	54.7	47.5			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% MCWB	40.5	39.2	46.6	55.7	66.3	67.8	71.4	68.2	61.5	56.5	52.0	45.8			
	2% DB	38.5	39.3	52.0	68.1	78.3	81.6	85.7	84.3	73.9	61.7	49.7	43.4			
	2% MCWB	36.7	37.1	43.0	53.3	63.4	67.1	69.1	68.8	60.9	55.6	47.7	42.0			
	5% DB	36.2	37.0	47.2	63.8	74.1	78.4	81.9	79.9	69.0	57.2	47.2	39.5			
	5% MCWB	34.7	35.3	40.8	51.6	60.7	65.8	67.6	65.8	58.6	52.5	45.4	38.5			
	10% DB	34.1	35.2	43.0	59.6	70.2	74.8	78.6	76.0	65.4	53.7	44.7	36.6			
	10% MCWB	33.0	33.8	38.4	49.6	58.5	63.5	66.3	63.9	56.9	50.1	43.2	35.4			
	0.4% WB	40.7	40.6	48.3	57.8	68.5	70.2	72.9	71.9	64.6	60.4	52.3	46.4			
	0.4% MCDB	41.9	42.3	57.1	69.3	79.9	80.3	84.9	82.0	76.1	66.0	54.3	47.4			
	2% WB	37.0	37.4	44.2	54.8	64.9	68.5	71.1	69.8	62.1	56.2	48.3	42.0			
	2% MCDB	38.1	38.6	50.7	64.2	74.8	78.9	82.7	81.2	70.6	60.5	49.4	43.1			
	5% WB	34.9	35.6	41.4	52.9	62.6	67.0	69.0	67.9	60.0	53.1	45.7	38.6			
	5% MCDB	35.9	36.8	46.2	61.5	72.3	76.5	78.6	78.8	66.4	56.4	46.9	39.5			
	10% WB	33.1	34.0	38.7	50.8	60.2	65.0	67.3	65.4	58.0	50.8	43.4	35.6			
	10% MCDB	34.1	35.0	42.5	58.3	68.3	73.4	76.2	73.5	63.2	53.6	44.6	36.5			
(35) Mean Daily Temperature Range	MDBR	8.3	10.2	12.3	16.6	18.4	17.7	18.3	19.2	17.2	12.4	7.5	7.4			
	5% DB MCDBR	6.0	9.0	18.3	23.3	24.2	22.4	24.0	25.8	23.7	18.3	9.9	7.6			
	5% DB MCWBR	5.9	7.7	11.0	11.5	11.6	10.6	10.2	10.9	12.1	11.9	8.1	6.9			
	5% WB MCDBR	5.8	8.3	16.2	20.3	21.9	20.4	21.0	22.9	20.0	16.1	8.5	7.5			
	5% WB MCWBR	5.9	7.3	10.6	10.9	11.2	10.7	10.2	11.1	11.4	11.6	7.6	7.0			
(40) Clear-Sky Solar Irradiance	taub	0.294	0.328	0.356	0.421	0.398	0.383	0.415	0.425	0.380	0.354	0.334	0.298			
	taud	2.132	2.070	2.123	2.143	2.295	2.357	2.288	2.245	2.337	2.374	2.257	2.160			
	Ebn at Noon	206	235	257	255	270	275	263	252	250	230	192	180			
	Edh at Noon	22	32	38	43	39	37	39	38	31	24	20	18			
(44) All-Sky Solar Radiation	RadAvg	233	516	964	1292	1720	1831	1757	1478	987	496	187	140			
	RadStd	47	90	139	132	119	157	151	145	97	97	36	37			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	+2.21	N/A	N/A	N/A	N/A
Regional (9 neighbors)	+1.12	N/A	N/A	N/A	N/A	N/A	-220	-355	+169	N/A

Nomenclature: See separate page